Aircraft Noise: Audibility-Based Labels and Day-Wise Validation

DESE71003 Sensing & IoT coursework · second training round · data 22 Oct – 31 Oct 2026 · written 1 Nov 2026 · follows the first training summary

1. Why this round

The first round labelled windows with fixed distances: **aircraft** = a pass within 3 km, **quiet** = nothing within 8 km, with everything in between left out. Those cut-offs were arbitrary. Planes are not equally audible at equal distances: some close ones are quiet and some distant ones are clearly heard. The first round also had two evaluation weaknesses: 86% of quiet windows were at night, and the model was tested on a single held-out day at the end of the week. This round (a) measures audibility directly and labels with it, and (b) replaces the single split with a mixed split and day-wise cross-validation.

2. Audibility experiment (analysis/audibility.py)

- **Tracks to 20 km.** The passes table only reaches about 5 km, but aircraft positions are logged out to 60 km. Every airborne track within 20 km is reduced to its closest approach: time, slant distance, bearing from the house and emitter category. That gives 12,452 tracks. 4,489 of them are *isolated*: no other aircraft within 1.5× their distance for ±45 s. Only isolated tracks are used for fitting.
- **Rise statistic.** The rise is the peak 5 s LAeq when the sound should arrive (closest time + slant / 343 m/s, window –10 to +15 s), minus the background LA90 of the previous 3 minutes. The 5 s mean ignores door slams. LA90 (ISO 1996) avoids the background being inflated by the previous arrival, which a median is not when arrivals come every 90 s; switching to it raised the measured audibility of overhead passes noticeably.
- **Why a plain dB threshold fails.** At random aircraft-free moments at 3 am, 13–19% still show a rise of over 15 dB. That is street traffic, which ADS-B cannot see. By day, the median aircraft-free rise is about 15 dB, from household activity. A rise alone therefore cannot say "heard".
- **Audibility score.** Each track's rise is compared with a *null*: the same statistic at random aircraft-free moments in the same period (day 07–23 or night). The score is the share of null rises the track beats. Averaged over a group, this is the ROC AUC of "aircraft vs nothing" from level alone: 0.5 means indistinguishable from no aircraft, 1.0 means always louder. Because it ignores spectral shape, it is a lower bound on what a person, or the octave-band classifier, can pick out.

Measured audibility score, isolated fixed-wing tracks (mic A; mic B within ±0.03)

Slant distance	Night (23–07)	Day (07–23)
~1.1 km, approach path south of house	0.86	0.53
~1.3 km, approach path north of house	0.94	0.87
4–6 km	0.74	0.55
6–8 km	0.62	0.51
8–10 km	0.68	0.55
10–13 km	0.67	–
13–20 km	0.56–0.57	–

Findings. (1) **Time of day matters more than distance.** At night aircraft stand out to about 13 km and fade by 15–20 km. By day, anything beyond about 3 km is indistinguishable from household noise by level. (2) **Direction matters more than a few hundred metres of distance.** Both Heathrow approach paths pass at about 3,100 ft heading west: one 0.56 km south of the house, the other 0.86 km north. The northern one is markedly more audible despite being further away, perhaps because of building orientation or screening. (3) Heavies (A5) are slightly louder than A3 airliners.

Model. A fractional logistic regression is fitted per mic. It predicts the score from ADS-B and the clock only: log slant distance, day/night and their interaction, direction terms (north/slant², east/slant²), heavy and helicopter. Its predictions match the measured score within about ±0.03 in almost every distance bin (Figure 1).

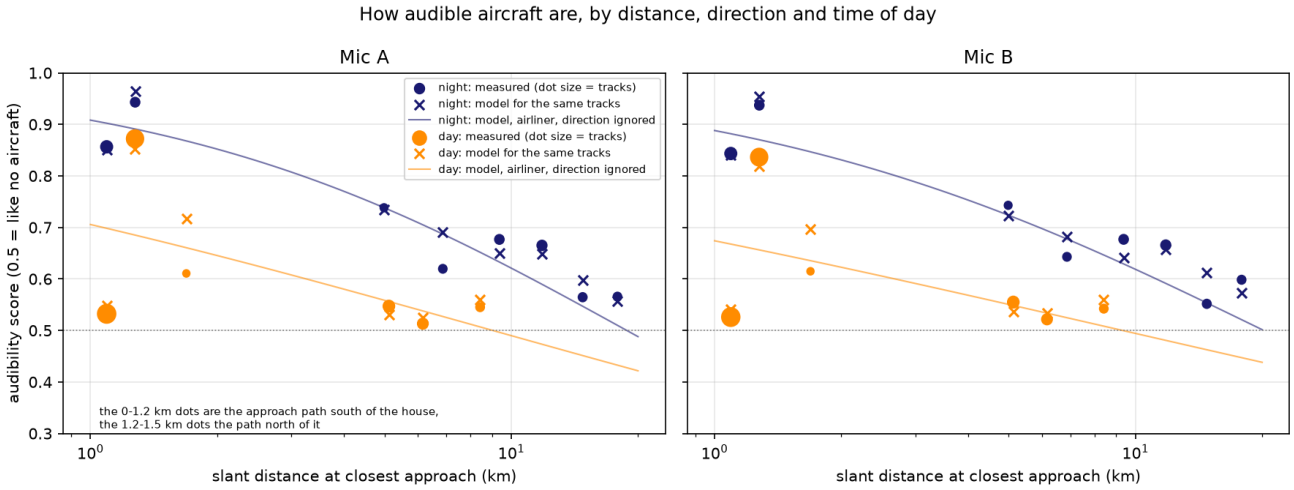


Figure 1. Measured audibility score (dots) and the model's prediction for the same tracks (crosses), by distance and time of day.

3. New labels (model/dataset.py)

- **Aircraft:** the window ends while the sound of a track with predicted audibility ≥ 0.8 for that mic is arriving.
- **Quiet:** every aircraft within 20 km for ±45 s scores ≤ 0.6, none is closer than 3 km, and the ADS-B logger is running. The 3 km floor is a safety margin, because the level-only score can miss close aircraft that are noticeable from their spectrum.
- **Excluded:** everything in between. This notably includes **daytime passes on the southern approach path** (score about 0.53). Whether a person hears them is unresolved; a short listening session would settle it.
- **Labels never come from a window's own sound.** A "loud = aircraft" label would teach the model that street traffic is aircraft.

Effect on the dataset (22–31 Oct): 95,451 windows, of which 42,175 are aircraft and 53,276 quiet. The night share of quiet windows fell from 86% to 53%. By day, a 5 km radius instead of 8 km would leave about 2.5× more daytime quiet candidates (13,904 vs 5,669). The aircraft class shrank because ambiguous passes are now left out.

4. Comparison on a fixed split (train 23–27, validation 28, test 29 Oct)

Test set	CNN trained on	AUC (daytime)	Recall	False alarms day / night
Old fixed-km test set	fixed-km labels	0.991 (0.975)	0.98	22.2% / 1.1%
Old fixed-km test set	audibility labels	0.987 (0.967)	0.96	15.4% / 1.0%
New audibility test set	fixed-km labels	0.979 (0.897)	0.97	21.9% / 1.2%
New audibility test set	audibility labels	0.977 (0.869)	0.95	14.3% / 1.2%

Audibility labels cut daytime false alarms by about a third, for about 2 points of recall, on either test set. AUC is unchanged, so the model's operating point moved rather than its underlying discrimination improving. The new test set's daytime AUC (about 0.87–0.90, against 0.975) shows that the old evaluation was optimistic: it excluded the hard daytime cases.

5. Mixed split and day-wise cross-validation (model/train.py)

- Days are classed as **busy** (westerly operations, about 65–75% aircraft windows: 23–28 Oct) or **quiet** (easterly operations, under about 20%: 29–31 Oct). Part days (22 Oct, which starts at 21:30) are used for training only.
- **Mixed split (default):** one random busy day and one random quiet day each for test and validation; training gets everything else, including recent days.
- **--cv:** each full day is the test day once, with validation on a random busy and quiet day from the rest. Reported as mean ± standard deviation over 9 folds. Whole days only, because of autocorrelation.

Detector	AUC	Daytime AUC	False alarms by day	False alarms at night
Threshold rule	0.81 ± 0.09	0.76 ± 0.12	40.6% ± 9.1%	23.0% ± 7.8%
Gradient boosting	0.97 ± 0.06	0.94 ± 0.07	8.9% ± 5.6%	1.4% ± 1.5%
1D CNN	0.97 ± 0.05	0.96 ± 0.06	5.8% ± 4.2%	0.6% ± 0.8%

CNN test day	23	24	25	26	27	28	29 (q)	30 (q)	31 (q)
Recall	0.89	0.99	0.96	0.97	0.97	0.96	0.85	0.23*	0.27*
False alarms	5.6%	3.4%	0.6%	0.6%	1.5%	0.7%	2.1%	2.3%	1.9%

(q) = quiet day. *Only 231 and 22 aircraft windows respectively.

Kept model (mixed split, seed 0: test 28 and 30 Oct, validation 25 and 29 Oct): precision 0.92, recall 0.95, F1 0.93, with false alarms on 3.8% of quiet seconds, against 9.1% in round 1. It has 13,281 parameters and trains in about 4 minutes on a laptop CPU.

6. Conclusions and next steps

- Training across all days, including the quiet ones, together with audibility labels cut daytime false alarms from 16–22% to about 6% on average, and under 1.5% on most busy days. Westerly arrivals are reliably detected (recall 0.89–0.99).
- **Weak spot:** the few aircraft heard on easterly days (recall about 25%). This is probably different traffic (departures, other routes, helicopters), with only about 250 training windows. More easterly days are the fix, so collection continues.
- Gradient boosting is close behind the CNN. This is consistent with the lecture's point that classical methods often perform surprisingly well.
- **Open items:** the labels are sensitive to the 0.8 cut-off, because daytime northern-path passes score 0.82–0.85; test 0.75. Choose the operating threshold for a target false-alarm rate rather than maximum F1. Add an event-level, continuous-stream test (false events per hour). Hold a listening session to tie the audibility score to human hearing. Try two-mic fusion.
- **Limitations:** 9 test days; easterly-day recall rests on few examples; the audibility score is level-only, a lower bound; the direction terms are effectively fitted from two approach paths.